Task 1: Intro to AI & Python - Study Notes

1. Basics of AI & Machine Learning

- **Artificial Intelligence (AI):** A foundational computer science concept for systems designed to exhibit intelligent behavior and understand context.
- **Machine Learning (ML):** A core concept under AI that relies on training data to function, allowing systems to find patterns and make inferences.

2. Difference Between AI, ML, and Deep Learning

- **AI:** The broadest category for all systems mimicking human intelligence.
- **ML:** A subset of AI focused on training models using data to make autonomous decisions without explicit programming.
- **Deep Learning:** A specialized ML subset using complex neural networks to process vast amounts of unstructured data.

3. Real-World AI Applications

- **Healthcare:** Smartphone tools identifying malnutrition in children.
- **Law Enforcement:** Tools helping investigators uncover new case leads.
- **Retail:** Companies like IKEA letting customers digitally redesign rooms.
- **Content Generation:** Platforms like Synthesia (avatars) and Descript (voice cloning).
- **Software Engineering:** Generative AI assisting with coding, debugging, and documentation.

4. Python Programming Concepts

- **Installation:** Download from python.org and pair with an IDE like VS Code.
- **Variables:** Named storage for values (strings, integers, floats). E.g., `name = "Alice"`.
- **Lists:** Ordered, changeable collections in square brackets. E.g., `fruits = ["apple", "banana"]`.
- **Loops:** Repeats a block of code. A `for` loop easily iterates through lists or numerical ranges.
- **Functions:** Reusable logic blocks using a `return` statement to send results back.

5. Beginner-Friendly Python Programs

- **Hello World:** `print("Hello, World!")` to output text.
- **Basic Arithmetic:** `sum = num1 + num2` for simple calculations.
- **Even or Odd Checker:** Using `input()`, `if/else`, and the modulus operator (`% 2 == 0`) to